Decentralized μ^2-SGD: Narrowing the Parallelism Gap to Centralized Learning

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Abstract

Decentralized learning has emerged as a scalable paradigm for training models across multiple machines while reducing communication overhead. However, increasing the number of machines beyond a certain point often negatively impacts convergence rates, thus substantially limiting scalability compared to centralized systems. In this work, we propose *Decentralized μ^2-SGD (DMS)*, a novel decentralized optimization algorithm that significantly extends the parallelism limits of decentralized learning. From a theoretical perspective, within the stochastic convex optimization (SCO) framework, we establish improved bounds on the maximal allowable parallelism, surpassing existing decentralized algorithms. Notably, for a broad class of network topologies, our method matches the parallelism scaling of centralized learning, thereby effectively eliminating the gap between decentralized and centralized optimization. Empirically, we validate our theoretical findings through extensive experiments, demonstrating the benefits of our approach.

1. Introduction

Distributed learning has emerged as a central paradigm for large-scale machine learning (ML), enabling multiple machines to collaboratively train a model by leveraging their aggregated data and computational capabilities (Verbraeken et al., 2020). This paradigm facilitates scalable ML by accelerating training via parallel computation. Distributed ML systems are particularly valuable when data is naturally dispersed across many users, as in federated learning (McMahan et al., 2017; Kairouz et al., 2021), where maintaining local data privacy is a primary concern.

Distributed learning systems are commonly classified into two main architectures: *centralized* and *decentralized*. Centralized approaches rely on a central parameter server (PS) to coordinate the training process by aggregating local updates and broadcasting the global model parameters (Li et al., 2014). Although this architecture simplifies training management, it suffers from scalability limitations caused by communication bottlenecks and is intrinsically susceptible to a single point of failure, as the server is critical to overall system operation.

In contrast, decentralized systems operate through direct interactions among neighboring machines, thereby eliminating the need for a central PS (Kempe et al., 2003; Boyd et al., 2006; Nedic & Ozdaglar, 2009; Lian et al., 2017). Although this architecture poses challenges in coordination and in preserving global model consistency, it also confers notable advantages. By mitigating the risk of a single point of failure and improving communication efficiency, decentralized systems offer increased flexibility and robustness (Assran et al., 2019; Kong et al., 2021; Yuan et al., 2021; Li et al., 2022).

A fundamental trade-off in distributed learning, in both centralized and decentralized settings, concerns the balance between parallelism and statistical efficiency. Although increasing the number of machines can accelerate training by enhancing parallelism, excessive parallelism might ultimately impair learning efficiency. This effect is particularly pronounced in decentralized systems, where sparse communication topologies amplify divergence among local models.

Existing decentralized methods exhibit a persistent gap relative to centralized learning in their ability to scale with the number of machines (Koloskova et al., 2020; 2021; He et al., 2022; Eisen et al., 2025). Notably, this gap remains even in highly connected network topologies, where rapid information propagation and improved mixing would, in principle, be expected to drive the dynamics closer to those of centralized training. This naturally raises the question of *whether decentralized methods are fundamentally limited in their ability to fully capitalize on increased parallelism compared to centralized approaches*.

In this work, we propose *Decentralized μ^2-SGD (DMS)*, a novel and simple algorithm designed to enhance parallelism in decentralized systems. Our method builds on the μ^2-SGD framework (Dahan & Levy, 2025) and enables decentralized learners to better exploit network connectivity by leveraging

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Table 1. **Parallelism Scaling.** Comparison of parallelism bounds for decentralized SCO across network topologies, expressed in terms of total sample count $N = MT$. We compare D-SGD Koloskova et al., 2020, DAT-SGD (Eisen et al., 2025), and our method against the centralized baseline of $\mathcal{O}(\sqrt{N})$. Well-connected (WC) graphs satisfy $\rho = \Omega(M^{-1/3})$; more generally, we parameterize connectivity as $\rho = \Omega(M^{-a})$ for $a \in [0, 1/3]$. Higher parallelism bounds indicate improved scalability without sacrificing statistical efficiency.

TOPOLOGY $(1/\rho)$	D-SGD	DAT-SGD	DMS
RING $\mathcal{O}(M^2)$	$\mathcal{O}(N^{1/8})$	$\mathcal{O}(N^{1/6})$	$\mathcal{O}(N^{3/16})$
2D TORUS $\mathcal{O}(M)$	$\mathcal{O}(N^{1/6})$	$\mathcal{O}(N^{1/4})$	$\mathcal{O}(N^{3/10})$
3D TORUS $\mathcal{O}(M^{2/3})$	$\mathcal{O}(N^{3/16})$	$\mathcal{O}(N^{3/10})$	$\mathcal{O}(N^{3/8})$
WC $\mathcal{O}(M^a)$	$\mathcal{O}(N^{1/(4+2a)})$	$\mathcal{O}(N^{1/(2+2a)})$	$\mathcal{O}(\sqrt{N})$

slowly evolving averaged iterates and reduced-variance gradient estimates. Our approach mitigates the bias induced by local model discrepancies, often referred to as the *consensus distance*, and leads to more stable and efficient convergence.

We conduct our analysis within the stochastic convex optimization (SCO) framework, a powerful setting that encompasses several classical problems such as linear regression, and also serves as a testbed for the design and analysis of more sophisticated ML algorithms.

Contributions We demonstrate that the performance gap between centralized and decentralized learning is not fundamental, and can be eliminated under appropriate algorithmic design and connectivity conditions. Our main contributions are as follows:

- We propose DMS , a novel decentralized optimization algorithm, and analyze its convergence rate within the stochastic convex optimization (SCO) framework.
- We establish the achievable parallelism scales as $\mathcal{O}\big(\min\{\sqrt{N},\, \rho^{3/2}N^{3/4}\}\big)$, improving upon existing bounds for a broad class of network topologies, including rings, d-dimensional tori, and exponential graphs.
- For sufficiently well-connected networks, specifically for all topologies satisfying $\rho \geq \mathcal{O}(M^{-1/3})$, *our method matches the parallelism scaling of centralized systems* $\mathcal{O}(\sqrt{N})$; see Corollary B.2.

1.1. Related Work

Centralized learning systems are widely adopted in both industry and academia, thanks to their simplicity and the effectiveness of Minibatch SGD (Dekel et al., 2012), which is the most widespread approach to training in such systems. While alternative centralized methods like local-SGD have been explored in recent years (McMahan et al., 2017; Kairouz et al., 2021), Minibatch SGD still serves as a cornerstone in large-scale training due to its ease of implementation and strong empirical performance.

Despite the enduring popularity of centralized training, decentralized systems present an appealing alternative by eliminating the need for a central parameter server and enabling direct communication among neighboring machines (Tsitsiklis, 1984; Kempe et al., 2003; Boyd et al., 2006; Nedic & Ozdaglar, 2009; Lian et al., 2017). This design not only mitigates the risk of a single point of failure but also offers greater scalability and flexibility (Assran et al., 2019; Kong et al., 2021; Yuan et al., 2021). A common mechanism underpinning these benefits is the *gossip averaging* protocol (Xiao & Boyd, 2004; Boyd et al., 2006), a low-overhead communication method that efficiently disseminates model updates across the network.

Decentralized training has been extensively studied in both convex and non-convex stochastic optimization. In the convex setting, Lian et al. (2017) introduced Decentralized-SGD (D-SGD), which was later extended by Koloskova et al. (2020) to handle time-varying communication topologies and randomized gossip protocols. However, these methods suffer in heterogeneous data regimes, a limitation addressed by Koloskova et al. (2021) through gradient tracking mechanism (Di Lorenzo & Scutari, 2016; Nedic et al., 2017; Pu & Nedić, 2021), which recovers the homogeneous-case convergence guarantees. Similar guarantees have also been established in the non-convex setting (Lian et al., 2017; Koloskova et al., 2020; 2021).

Notably, existing convergence bounds reveal a persistent gap between centralized and decentralized learning, even in highly connected network topologies. While centralized methods support up to $\mathcal{O}(\sqrt{N})$ workers without degrading generalization (Dekel et al., 2012), the best known decentralized methods are limited to $M \leq \mathcal{O}\big((\rho N)^{3/8}\big)$ in the non-convex setting (Huang et al., 2024), and to $M \leq \mathcal{O}(\rho\sqrt{N})$ in the convex setting (Eisen et al., 2025). We provide a detailed comparison across network topologies in Table 1.

To further improve the parallelism bound, we extend the rationale of Eisen et al. (2025) by adopting a recent technique from SCO that incorporates gradually shifting query points together with reduced-variance gradient estimates (Cutkosky & Orabona, 2019; Dahan & Levy, 2025). This approach has recently been used to improve privacy in federated learning (Reshef & Levy, 2025) and robustness in asynchronous learning (Dahan & Levy, 2024).

2. Problem Setup and Preliminaries

In this section, we introduce the problem setting and define the notation and assumptions used throughout the paper.
We study decentralized SCO, where the objective is to mini-

Table 2. Comparison of convergence rates to ϵ-Accuracy in convex decentralized learning for different spectral gaps ρ, data heterogeneity ζ, and noise variance σ. The dominant term $\sigma^2/(M\epsilon^2)$ matches the statistically optimal rate (Nemirovski & Yudin, 1983).

REFERENCE	CONVERGENCE RATE
KOLOSKOVA ET AL. (2021)	$\mathcal{O}\left(\frac{\sigma^2}{M\epsilon^2} + \frac{\sigma}{\sqrt{\rho}\epsilon^{3/2}} + \frac{1}{\rho\epsilon}\right)$
EISEN ET AL. (2025)	$\mathcal{O}\left(\frac{\sigma^2}{M\epsilon^2} + \frac{\sqrt{\sigma}+\sqrt{\zeta}}{\rho\epsilon} + \frac{1}{\epsilon}\right)$
DMS (OURS)	$\mathcal{O}\left(\frac{\sigma^2}{M\epsilon^2} + \left(\frac{1+\zeta^2}{M\rho^6\epsilon^4}\right)^{1/5} + \frac{1}{\sqrt{M\epsilon^3}}\right)$

mize a convex function $f : \mathbb{R}^d \to \mathbb{R}$ defined as

$$f(x) := \frac{1}{M}\sum_{i=1}^{M} \{f_i(x) := \mathbb{E}_{z\sim\mathcal{D}_i}[f_i(x,z)]\} , \qquad (1)$$

where $f_i : \mathbb{R}^d \to \mathbb{R}$ denotes the local objective at machine $i \in \{1,\dots,M\}$, $\mathcal{D}_i$ is the corresponding data distribution, and $f_i(\cdot, z)$ denotes the loss associated with data point z on machine i.

We consider first-order optimization algorithms. At iteration t, each machine i draws a sample $z_t^i \sim \mathcal{D}_i$, computes a stochastic gradient estimate $\nabla f_i(x_t^i, z_t^i)$, and updates its iterate x_t^i. The algorithm outputs an estimate x_{output}, whose performance is evaluated using the expected *excess loss*,

$$\mathbb{E}[\text{Excess-Loss}] := \mathbb{E}[f(x_{\text{output}})] - f(x^*) ,$$

where $x^* \in \arg\min_{x\in\mathbb{R}^d} f(x)$ denotes the global minimizer, and the expectation is taken over the randomness of the samples.

Throughout the paper, we adopt the following assumptions.

Assumption 2.1 (Smoothness)**.** For all $i \in [M]$, the function f_i is L-smooth; that is, for all $x, y \in \mathbb{R}^d$, $z \in$ **Sup**$\{\mathcal{D}_i\}$,

$$\|\nabla f_i(x;z) - \nabla f_i(y;z)\| \le L\|x-y\|.$$

Assumption 2.2 (Bounded noise variance)**.** For all $i \in [M]$ and all $x \in \mathbb{R}^d$, there exists a constant σ^2 such that

$$\mathbb{E}_{z\sim\mathcal{D}_i}\left[\|\nabla f_i(x,z) - \nabla f_i(x)\|^2\right] \le \sigma^2 .$$

Assumption 2.3 (Bounded smoothness variance (Dahan & Levy, 2025))**.** Assumption 2.1 implies that for all $i \in [M]$ and all $x, y \in \mathbb{R}^d$, there exists a constant $\sigma_L^2 \in [0, L^2]$ such that

$$\mathbb{E}\left\|\nabla f_i(x;z) - \nabla f_i(x) - \big(\nabla f_i(y;z) - \nabla f_i(y)\big)\right\|^2 \le \sigma_L^2\, \mathbb{E}\|x-y\|^2 .$$

Assumption 2.4 (Bounded heterogeneity)**.** For all $x \in \mathbb{R}^d$, there exists a constant ζ^2 such that

$$\frac{1}{M}\sum_{i=1}^{M} \|\nabla f_i(x) - \nabla f(x)\|^2 \le \zeta^2 .$$

The bounded heterogeneity assumption is widely adopted in prior work (Lian et al., 2017; Tang et al., 2018; Li et al., 2019; He et al., 2022), although milder variants have been studied, requiring bounded heterogeneity only at the optimum (Koloskova et al., 2020) or at the origin (Lu & De Sa, 2021). For simplicity, we adopt the standard assumption.

Decentralized Training. We consider distributed systems in which computational nodes are connected by a communication graph, enabling direct information exchange among neighbors. In this decentralized setting, nodes correspond to individual machines, and edges represent available communication links. For inter-machine communication, we adopt *gossip averaging*, a widely used and robust protocol that enables efficient information propagation across the network without relying on a central coordinator (Xiao & Boyd, 2004; Boyd et al., 2006; Nedic & Ozdaglar, 2009; Shi et al., 2014; Koloskova et al., 2020; Eisen et al., 2025).

Definition 2.5 (Gossip matrix)**.** A *gossip matrix* $P \in [0,1]^{M\times M}$ is a symmetric ($P = P^\top$), doubly stochastic ($P\mathbf{1} = \mathbf{1}$, $\mathbf{1}^\top P = \mathbf{1}^\top$) matrix, where $P_{ij} > 0$ if and only if nodes i and j are connected.

Concretely, suppose each machine i is initialized with a vector $x_0^i \in \mathbb{R}^d$, and the objective is to propagate the *consensus* value $\bar{x} := \frac{1}{M}\sum_i x_0^i$ across the network. Gossip averaging provides an iterative procedure whereby each machine generates a sequence $\{x_t^i\}_t$ converging to $\bar{x}$. At each step, node i updates its local vector by forming a weighted combination of its neighbors' values as follows:

$$x_{t+1}^i = \sum_{j=1}^{M} P_{ij} x_t^j ,$$

The gossip update can be expressed in matrix form as

$$X_{t+1} = X_t P ,$$

where $X_t := \begin{pmatrix} x_t^1 & x_t^2 & \cdots & x_t^M \end{pmatrix} \in \mathbb{R}^{d\times M}$. This iterative averaging operation preserves the global consensus, namely, for all t, $\bar{x} = \frac{1}{M}\sum_{i=1}^M x_0^i = \frac{1}{M}\sum_{i=1}^M x_t^i$.

Throughout this work, we assume that the communication graph is connected and that the mixing matrix P is fixed and satisfies the properties stated above.

We next introduce a key characteristic of the gossip matrix, namely the *spectral gap*, which quantifies the connectivity of the underlying network topology.

Definition 2.6 (Spectral gap)**.** Let P be a gossip matrix with eigenvalues $1 = \lambda_1 > |\lambda_2| \geq \cdots \geq |\lambda_M|$. The *spectral gap* of P is defined as

$$\rho \coloneqq 1 - |\lambda_2| \in (0, 1].$$

Many deterministic graphs with geometric topologies (e.g., ring, d-dimensional torus (Nedić et al., 2018)) exhibit a spectral gap that scales as $\rho = \mathcal{O}(M^{-a})$ for some $a > 0$ (see Table 1), while $\rho = 1$ corresponds to the centralized topology (complete graph). This scaling behavior will be explicitly exploited in our analysis.

A fundamental property of gossip averaging is its contraction of local deviations from the consensus value $\bar{x}$.

Property 2.7. Let $P \in \mathbb{R}^{M\times M}$ be a gossip matrix with spectral gap ρ, and let $X \in \mathbb{R}^{d\times M}$ with $\bar{X} \coloneqq X\frac{1}{M}\mathbf{1}\mathbf{1}^\top$. Then,

$$\|XP - \bar{X}\|_F^2 \leq (1-\rho)\|X - \bar{X}\|_F^2 \,. \tag{2}$$

This contraction implies exponential convergence to consensus under repeated gossip updates; in particular, complete graphs reach consensus in a single iteration.

3. The Fragility of Consensus Bias

We begin by examining the fundamental performance gap between centralized and decentralized learning through the lens of achievable parallelism. We then introduce *consensus bias*, one of the core challenges in decentralized optimization that persists even in well-connected networks. Finally, we present DAT-SGD, a recent method that partially mitigates this bias and serves as the foundation for our approach.

3.1. Parallelism Bounds Gap

In distributed learning, computation is partitioned across M machines to accelerate training. Assuming each machine processes one sample per iteration, then N samples are processed in $T = N/M$ iterations. This parallelization introduces an inherent trade-off, as increasing the number of machines may eventually offset computational gains due to inefficiencies.

In the centralized setting, Dekel et al. (2012) characterized the convergence rate of mini-batch SGD as

$$\mathbb{E}[\text{Excess-Loss}] = \mathcal{O}\left(\frac{\sigma}{\sqrt{MT}} + \frac{1}{T}\right) = \mathcal{O}\left(\frac{\sigma}{\sqrt{N}} + \frac{M}{N}\right).$$

Consequently, up to $M = \mathcal{O}(\sqrt{N})$, and ignoring the dependence on σ, increased parallelism does not degrade statistical performance.

In decentralized optimization, the lack of global communication and exact synchronization introduces additional error beyond the statistical term $\sigma/\sqrt{MT}$, arising from consensus bias and limiting the achievable parallelism. Recent work (Eisen et al., 2025) showed that this limitation can be substantially mitigated by controlling the consensus error, yielding an improved regime with admissible scaling $M = \mathcal{O}(\rho\sqrt{N})$. While this suggests that decentralized methods can approach centralized behavior under strong connectivity, a nontrivial gap remains. Building on this insight, we seek to further extend the achievable parallelism while preserving statistical efficiency.

3.2. Consensus Bias in Decentralized Training

Consensus Bias is a fundamental limitation in decentralized optimization. To expose the underlying phenomenon, we consider Decentralized SGD (D-SGD), which is a natural extension of standard SGD to decentralized systems. At each iteration, each machine alternates between performing a local stochastic gradient update and exchanging information with its neighbors via gossip averaging, namely,

$$w^i_{t+\frac{1}{2}} = w^i_t - \eta\nabla f_i(w^i_t, z^i_t), \quad w^i_{t+1} = \sum_{j=1}^{M} w^j_{t+\frac{1}{2}} P_{ij}.$$

where w^i_t denotes the local model weights at node i and iteration t, and z^i_t the corresponding sample.

In contrast to centralized training, where all workers evaluate gradients at a common point, decentralized execution lacks immediate synchronization. Since gossip communication does not enforce instant consensus (unless $\rho = 1$), local models drift apart and nodes evolve independently.

This lack of communication introduces a central challenge: **local gradient estimates become biased relative to the global consensus**, which degrades the convergence rate. As shown in (Koloskova et al., 2019), this bias is intrinsically linked to the consensus distance $\Xi_t \coloneqq \frac{1}{M}\sum_{i=1}^{M}\|w^i_t - \bar{w}_t\|^2$, where $\bar{w}_t \coloneqq \frac{1}{M}\sum_{i=1}^{M} w^i_t$ denotes the consensus value and Ξ_t quantifies disagreement among local models. Limited communication restricts the attainable contraction rate, reflecting a trade-off between how quickly gossip reduces disagreement and how strongly local stochastic updates induce divergence.

3.3. Decentralized Anytime SGD (DAT-SGD)

The Anytime-SGD framework (Cutkosky, 2019) modifies standard SGD by decoupling the optimization iterate from the point at which gradients are queried. Instead of evaluating gradients at the current *model iterate* $\{w_t\}_t$, Anytime maintains a secondary sequence of *query points* $\{x_t\}_t$, defined recursively as weighted averages of past iterates with respect to non-negative weights $\{\alpha_t\}_t$ 4. This ensures that the query points converge to the optimal solution while

achieving the same convergence rates of SGD for convex objectives. Moreover, the query points evolve more smoothly than the raw iterates, exhibiting improved stability that enables effective control of consensus bias in decentralized training.

Eisen et al. (2025) extends this mechanism to decentralized optimization by coupling the local Anytime updates with gossip-based communication (5). The local averaging step stabilizes the evolution of the query points, while gossip communication gradually aligns the local variables across machines. Concretely, for node i at iteration t, the updates take the form,

$$w^i_{t+\frac{1}{2}} = w^i_t - \eta\, \alpha_t d_t, \tag{3}$$

$$x^i_{t+\frac{1}{2}} = \frac{\alpha_{1:t}}{\alpha_{1:t+1}}\, x^i_t + \frac{\alpha_{t+1}}{\alpha_{1:t+1}}\, w^i_{t+\frac{1}{2}}, \tag{4}$$

$$w^i_{t+1} = \sum_{j=1}^{M} P_{ij}\, w^j_{t+\frac{1}{2}}, \quad x^i_{t+1} = \sum_{j=1}^{M} P_{ij}\, x^j_{t+\frac{1}{2}}. \tag{5}$$

Where $d_t = \nabla f(x^i_t, z^i_t)$, $w_1 = x_1$, $\alpha_{1:t} := \sum_{\tau=1}^{t} \alpha_\tau$ for all $\tau \geq 1, \alpha_{1:0} := 0$. By propagating both the model parameters and the stabilized query points through the network, DAT-SGD mitigates drift between local iterates in decentralized settings. *This mechanism is the starting point for the developments in the present work.*

4. DMS : Decentralized μ^2-SGD

In this section, we introduce DMS . We begin in Section 4.1 by presenting the μ^2-SGD method, which forms the foundation of our approach. We then extend the framework to the decentralized setting and present the main convergence result in Section 4.2. Finally, Section 4.3 provides an intuitive analysis illustrating how DMS mitigates the consensus distance, followed by a proof sketch.

4.1. μ^2-SGD

The μ^2-SGD method (Dahan & Levy, 2025) extends standard SGD by combining the Anytime-SGD weighted averaging mechanism with variance reduction via *corrected momentum*. μ^2-SGD retains the Anytime-SGD weight dynamics from Equations (3) and (4) and replaces the stochastic gradient oracle with a stabilized momentum-based estimator. The corrected momentum estimator, inspired by the STORM algorithm (Cutkosky & Orabona, 2019), is defined as:

$$d_t = \nabla f(x_t, z_t) + (1-\beta_t)\left(d_{t-1} - \nabla f(x_{t-1}, z_t)\right), \tag{6}$$

where $\beta_t \in [0,1]$ denotes the corrected momentum weight. Although d_t is *conditionally* biased, i.e., $\mathbb{E}[d_t \mid x_t] \neq \nabla f(x_t)$, it remains unbiased under *unconditional* expectation: $\mathbb{E}[d_t] = \nabla f(x_t)$.

Algorithm 1 Decentralized μ^2-SGD (DMS)

Input: Number of rounds T, learning rate η, gossip matrix P, non-negative weights $\{\alpha_t\}_{t\geq 1}$, corrected momentum weights $\{\beta_t\}_{t\geq 1}$, initial point x_1.
Initialize: For each $i \in [M]$, set $w^i_1 = x^i_1 = x_1$. Draw a sample $z^i_1 \sim \mathcal{D}_i$ and define $d^i_1 = \nabla f_i(x^i_1, z^i_1)$.
for $t = 1, \ldots, T$ **do**
  for $i \in [M]$**(in parallel) do**
    Local Updates
    $w^i_{t+\frac{1}{2}} = w^i_t - \eta\alpha_t d^i_t$
    $x^i_{t+\frac{1}{2}} = \frac{\alpha_{1:t}}{\alpha_{1:t+1}} x^i_t + \frac{\alpha_{t+1}}{\alpha_{1:t+1}} w^i_{t+\frac{1}{2}}$
    Weights Gossip Communication
    $w^i_{t+1} = \sum_{j=1}^{M} w^j_{t+\frac{1}{2}} P_{ij}$
    $x^i_{t+1} = \sum_{j=1}^{M} x^j_{t+\frac{1}{2}} P_{ij}$
    Local Gradients Evaluation
    Sample $z^i_{t+1} \sim \mathcal{D}_i$, and compute:
    $g^i_{t+1} = \nabla f_i(x_{t+1}, z_{t+1})$, $\tilde{g}^i_{t+\frac{1}{2}} = \nabla f_i(x_t, z_{t+1})$.
    Corrected Momentum Estimates
    $d^i_{t+\frac{1}{2}} = g^i_{t+1} + (1-\beta_{t+1})\left(d^i_t - \tilde{g}^i_{t+1}\right)$
    $d^i_{t+1} = \sum_{j=1}^{M} d^j_{t+\frac{1}{2}} P_{ij}$
  end for
end for

The estimator enjoys two key properties that enable improved convergence guarantees under bounded domain assumptions: (i) *diminishing variance*: with $\beta_t = 1/t$, the estimator satisfies $\mathbb{E}\left\|d_t - \nabla f(x_t)\right\|^2 \leq \mathcal{O}\left(\frac{1}{t}\left(\sigma^2 + \zeta^2\right)\right)$ (Dahan & Levy, 2025), in contrast to the uniform bound $\mathcal{O}(\sigma^2 + \zeta^2)$ for standard SGD; and (ii) *strong temporal correlation*: successive gradient estimates satisfy $\left\|d_t - d_{t-1}\right\|^2 \leq \mathcal{O}\left(S^2/t^2\right)$, where $S^2 = \mathcal{O}(1)$ is independent of t, inducing smooth gradient dynamics over time (see Proposition H.2).

4.2. Algorithm and Main Result

We extend μ^2-SGD to the decentralized regime in Algorithm 1. Building on the methodology of Eisen et al. (2025), the resulting procedure retains the structure of DAT-SGD while incorporating corrected momentum for variance reduction.

Each iteration t of DMS (Algorithm 1) leverages two complementary stabilization mechanisms: (i) Anytime-SGD regularizes the evolution of the iterates through local updates of the model variables and query points according to Equations (3) and (4), followed by gossip communication to synchronize the model weights w^i_t and query points x^i_t across neighboring nodes (5); and (ii) stabilized μ^2 gradient estimates, benefiting from reduced variance and enhanced temporal stability, are propagated via gossip to suppress gradient-induced variability across the network.

The convergence of Algorithm 1 is established in Theorem 4.1, with the full proof deferred to Section B.

Theorem 4.1. *Under Assumptions 2.1-2.4, consider Algorithm 1 with linear weights $\alpha_t = t$, momentum weights $\beta_t = 1/t$ and a learning rate given by*

$$\eta = \frac{1}{L}\min\left\{\frac{\rho^3}{464}, \frac{\left(M\rho^6\right)^{1/4}}{146\,T^{3/4}}, \frac{1}{32T}, \frac{M^{1/3}}{72T^{4/3}}, \frac{\sqrt{M}}{\sqrt{2688}T^{3/2}}\right\}.$$

Then, for all $T \geq 1$, the following bound holds:

$$\mathbb{E}[f(\bar{x}_T) - f^\star] \leq \mathcal{O}\Bigg(\frac{\mathcal{K}_1\sigma + \mathcal{K}_1^2 L}{\sqrt{MT}} + \frac{\mathcal{K}_1^2\, L}{M^{1/3}T^{2/3}} + \frac{\mathcal{K}_1^2\, L}{T} + \frac{\mathcal{K}_1^2\, L + \mathcal{K}_1^{3/2}\sqrt{L\zeta}}{M^{1/4}\rho^{3/2}T^{5/4}} + \frac{\mathcal{K}_1^2 L}{\rho^3 T^2}\Bigg),$$

where $\bar{x}_T := \frac{1}{M}\sum_{i=1}^M x_T^i$ and $\mathcal{K}_1^2 := \|w_1 - x^*\|^2$.

In terms of total sample complexity $N = MT$, and ignoring dependencies on L, $\mathcal{K}_1$, σ, and ζ, Lemma B.1 shows that the achievable asymptotic upper bound on parallelism scales as $\mathcal{O}\big(\min\{\sqrt{N},\, \rho^{3/2}N^{3/4}\}\big)$. Increasing the number of machines beyond this regime leads to a degradation in learning efficiency. This substantially improves over the prior $\mathcal{O}(\rho\sqrt{N})$ limit (Eisen et al., 2025) for a broad class of network topologies. Moreover, for sufficiently well-connected graphs, DMS attains centralized-level parallelism in a fully decentralized setting.

4.3. Intuition and Proof Sketch

Before presenting the proof sketch, we provide intuition for why DMS achieves improved convergence rates compared to prior decentralized methods.

Intuition. To build intuition, we analyze the dynamics under simplifying assumptions; a rigorous treatment is provided in Appendix E. We assume that $f(\cdot, z)$ is *G-Lipschitz* and *L-smooth*, and that the iterates lie in a bounded domain $\mathcal{C}$, i.e., $\|x - y\|^2 \leq \mathcal{K}^2$ for all $x, y \in \mathcal{C} \subset \mathbb{R}^d$.

As shown by Reshef & Levy (2025), choosing $\alpha_t = 1/\beta_t = t$ yields an equivalent recursion for the gradient estimates in (6):

$$\begin{aligned} r_t &:= t\,\nabla f(x_t, z_t) - (t-1)\,\nabla f(x_{t-1}, z_t), \\ td_t &= (t-1)d_{t-1} + r_t. \end{aligned} \tag{7}$$

where r_t captures the gradient increments. Writing the recursion in matrix form and unrolling it yields

$$D_t = \frac{1}{t}D_1 P^{t-1} + \frac{1}{t}\sum_{\tau=1}^{t-2} R_{t-\tau}P^{t-\tau+1}. \tag{8}$$

As illustrated in (8), the momentum recursion accumulates the contraction induced by gossip averaging across iterations, thereby strengthening the effective mixing and leading to an inverse-linear decay of the gradient consensus bias. This representation implies the following bound (see Proposition H.3):

$$\|D_t - \bar{D}_t\|^2 \leq \mathcal{O}\left(\frac{MS^2}{\rho\, t}\right), \tag{9}$$

where $S^2 = \mathcal{O}(1)$ is a constant determined by the problem parameters, and $\|D_t - \bar{D}_t\|^2 := \sum_{i=1}^M \|d_t^i - \bar{d}_t\|^2$ denotes the *gradient estimate consensus distance*. In contrast, for a non-momentum-based approach the corresponding bound remains $\mathcal{O}(M)$, indicating the absence of decay over time.

As discussed in Section 3, controlling the consensus distance is essential for bias reduction. In particular, the consensus bias of the query points, $\Gamma_t := \frac{1}{M}\sum_{i=1}^M \|x_t^i - \bar{x}_t\|^2$, governs the convergence rate and satisfies

$$\mathcal{J} := \sum_{t=1}^T t^2\Gamma_t \leq \mathcal{O}\left(\frac{\eta^2}{M\rho^4}\sum_{t=1}^T t^2\|D_t - \bar{D}_t\|^2\right). \tag{10}$$

For DAT-SGD, a bound of the same form can be obtained. Substituting the estimate $\|D_t - \bar{D}_t\|^2 = \mathcal{O}(M)$ gives

$$\mathcal{J}_{\text{DAT}} \leq \mathcal{O}\left(\frac{\eta^2 T^3}{\rho^4}\right). \tag{11}$$

In contrast, plugging (9) into (10) gives

$$\mathcal{J}_{\text{DMS}} \leq \mathcal{O}\left(\frac{\eta^2 S^2 T^2}{\rho^5}\right),$$

and the precise analysis in Appendix E further tightens this bound to

$$\mathcal{J}_{\text{DMS}} \leq \mathcal{O}\left(\frac{\eta^2 S^2 T}{\rho^6}\right). \tag{12}$$

The proof sketch below derives the error bound for DMS ; for comparison, we note that the bounds for both methods admit the following structural form:

$$\text{Err}_{\text{DMS}} = \mathcal{O}\left(\frac{1}{\eta} + \frac{\eta T^3\sigma^2}{M} + \eta T\cdot\frac{T}{M}\mathcal{J}_{\text{DMS}}\right)\frac{1}{T^2},$$
$$\text{Err}_{\text{DAT}} = \mathcal{O}\left(\frac{1}{\eta} + \frac{\eta T^3\sigma^2}{M} + \eta T\cdot\mathcal{J}_{\text{DAT}}\right)\frac{1}{T^2},$$

where we suffer a factor of $T/M \geq 1$ over the accumulated consensus bias $\mathcal{J}$ compared to the analogous term in DAT-SGD analysis since our gradient estimator d_t is **conditionally biased**. This leads to an overall error rate of

$$Err_{\text{DMS}} = \mathcal{O}\left(\frac{1}{\eta T^2} + \frac{\eta T\sigma^2}{M} + \boldsymbol{\frac{\eta^3 T}{M\rho^6}}\right),$$
$$Err_{\text{DAT}} = \mathcal{O}\left(\frac{1}{\eta T^2} + \frac{\eta T\sigma^2}{M} + \boldsymbol{\frac{\eta^3 T^2}{\rho^4}}\right).$$

Optimizing the step size η recovers the rate stated in Theorem 4.1. Thus, the conditionally biased estimators reduce the consensus bias, achieving improved convergence despite the additional factor in bias propagation.

Proof Sketch. For simplicity, we retain only the leading-order terms in the expressions, omit numerical constants, and adopt the crude assumption that σ, ζ, L, and $\mathcal{K}_1$ satisfy $\mathcal{O}(1)$ bounds.

As established in Appendix B.2, the sequences $\{\bar{x}_t\}_{t\geq 1}$ and $\{\bar{w}_t\}_{t\geq 1}$ follow the same structural dynamics as Anytime-SGD, thereby enabling the use of standard guarantees from the Anytime framework. In particular, letting $\Delta_t := \mathbb{E}[f(\bar{x}_t) - f^*]$, we obtain for all $t \geq 1$ (Theorem A.1):

$$\alpha_{1:T}\Delta_T \lesssim \frac{1}{\eta} + \eta \sum_{\tau=1}^{T} \alpha_\tau^2 \mathbb{E}\|\bar{d}_\tau\|^2 + \eta T \cdot B_T \ , \tag{13}$$

where $B_T := \mathbb{E}\Big[\sum_{\tau=1}^{T} \alpha_\tau^2 \left\|\bar{d}_\tau - \nabla f(\bar{x}_\tau)\right\|^2\Big]$. Focusing on the middle term, we show using standard arguments that:

$$\sum_{\tau=1}^{T} \alpha_\tau^2 \mathbb{E}\left\|\bar{d}_\tau\right\|^2 \lesssim B_T + \sum_{\tau=1}^{T} \alpha_\tau^2 \mathbb{E}\left\|\nabla f(\bar{x}_\tau)\right\|^2 \ ,$$

which implies

$$\alpha_{1:T}\Delta_T \lesssim \frac{1}{\eta} + \eta \underbrace{\sum_{\tau=1}^{T} \alpha_\tau^2 \mathbb{E}\left\|\nabla f(\bar{x}_\tau)\right\|^2}_{\Lambda} + \eta T \cdot B_T, \tag{14}$$

Observe that the term Λ can be linearly expressed in terms of $\alpha_{1:T}\Delta_T$, tuning η we can therefore absorb it into the term on the left hand side and obtain:

$$\alpha_{1:T}\Delta_T \lesssim \frac{1}{\eta} + \eta T \cdot B_T \ , \tag{15}$$

A key step in the analysis is to control the term B_T, which quantifies the cumulative weighted divergence of the gradient estimates. We therefore turn to bounding B_T using Lemma C.1.

$$B_T \lesssim \sum_{\tau=1}^{T} \mathbb{E}\left\|\alpha_\tau \bar{\epsilon}_\tau\right\|^2 + \sum_{\tau=1}^{T} \alpha_\tau^2 \Gamma_\tau \ , \tag{16}$$

where $\bar{\varepsilon}_t := \bar{d}_t - \overline{\nabla f}(x_t)$ are the averaged estimate's error. From Lemma D.1 we can infer that upon setting $\alpha_\tau = \tau$, the estimate's error admits the following:

$$\sum_{t=1}^{T} \mathbb{E}\|t\bar{\varepsilon}_t\|^2 \lesssim \frac{\sigma^2 T^2}{M} + \frac{T}{M}\sum_{\tau=1}^{T} \tau^2 \Gamma_\tau \ , \tag{17}$$

Thus, combining Equation (17) into Equation (16) into (15) yields:

$$\alpha_{1:T}\Delta_T \lesssim \frac{1}{\eta} + \eta T \left(\frac{\sigma^2 T^2}{M} + \frac{T}{M}\sum_{\tau=1}^{T} \tau^2 \Gamma_\tau\right) \ . \tag{18}$$

As discussed earlier, the consensus-bias term carries an additional factor of T/M relative to DAT-SGD due to the conditional bias of d_t. Since we operate in the decentralized regime, the inequality $T/M \geq 1$ follows directly from the centralized parallelism bound $\mathcal{O}(\sqrt{N})$ together with the identity $N = MT$. However, the overall bound on the main term then follows as stated in the main theorem, improving upon DAT-SGD and, to the best of our knowledge, all prior work. As discussed earlier, this directly translates to improved parallelism: for a given error tolerance, more machines can be effectively utilized, yielding better time efficiency compared to existing methods. This proof sketch focuses on the dominant error contributions; the full treatment of all secondary terms is provided in Appendix B.

5. Experiments

We empirically validate our theoretical findings on a synthetic least-squares problem, addressing two key research questions: **(1)** Can decentralized methods achieve centralized-level performance on well-connected networks? and **(2)** How does network topology affect parallelism scalability? We examine the number of machines that can be effectively utilized before network-induced errors dominate, comparing our approach against DAT-SGD (Eisen et al., 2025), which achieves the best known parallelism bounds to our knowledge. A full-resolution version of the figures is provided in Appendix I.

5.1. Setup: Least Squares on Synthetic Data

We follow the experimental setting of (Koloskova et al., 2020; Eisen et al., 2025). Each machine's local objective is $f_i(x) = \frac{1}{2}\|A_i x - b_i\|^2$, with $A_i \in \mathbb{R}^{d\times d}$ drawn from a standard multivariate normal distribution. The target vector is set as $b_i = A_i(x^\sharp - \delta_i)$, where $x^\sharp \sim \mathcal{N}(0, \frac{1}{d}I_d)$ is sampled once per configuration, and $\delta_i \sim \mathcal{N}(0, \frac{\zeta^2}{d}I_d)$ introduces heterogeneity across machines.[1] To incorporate stochasticity, we add Gaussian noise $\xi \sim \mathcal{N}(0, \frac{\sigma^2}{d}I_d)$ when querying local gradients, resulting in the noisy gradient estimate $\nabla f_i(x) + \xi$. We set $d = 50$ in all experiments. All results are averaged over multiple random seeds (three for Erdős-Rényi graphs and five for fixed topologies).

Erdős-Rényi Communication Graphs: Approaching Centralized Performance The Erdős-Rényi graph model (Durrett, 2007) generates random graphs where each

[1] Here, ζ^2 quantifies heterogeneity **at the optimum**.

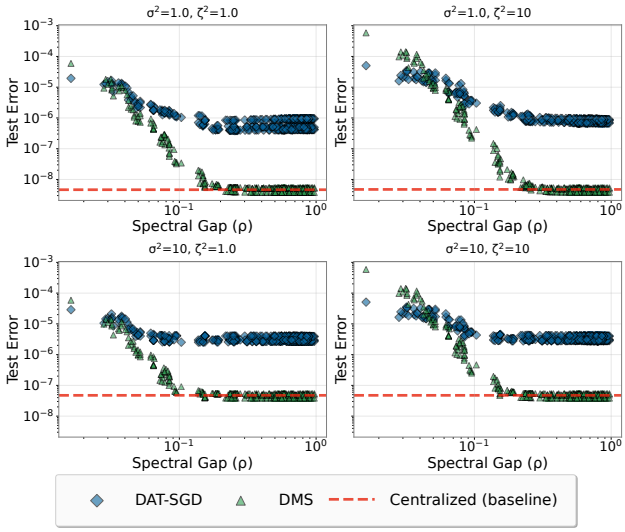

Figure 1. Comparison of DMS and DAT-SGD with $M = 200$ machines across varying network connectivity (ρ) for different noise variance (σ) and heterogeneity levels (ζ). The centralized baseline (dashed line) represents the performance with full connectivity ($\rho = 1$). DMS achieves near-centralized performance for $\rho \gtrsim 0.1$, substantially outperforming DAT-SGD across all connectivity regimes.

edge is independently included with a fixed probability. This flexible mechanism allows us to control network connectivity by tuning the edge probability.

We fix $M = 200$ machines and generate graphs using 200 edge probabilities, sampling 3 random graphs per probability to obtain a wide range of spectral gaps. In our samples, no graphs with spectral gap equal to 1 were observed. In Figure 1, we compare the performance of DMS and DAT-SGD across four configurations defined by $\sigma, \zeta \in \{1, 3.16\}$. The centralized baseline (dashed line) corresponds to an additional experiment where all machines communicate directly ($\rho = 1$).

DMS not only substantially outperforms DAT-SGD across all spectral gap regimes, but achieves near-centralized performance for a wide range of connectivity levels, particularly when $\rho \gtrsim 0.1$. This supports our main theoretical conclusion: *the performance gap between centralized and decentralized optimization is not fundamental*, and can be largely eliminated with improved consensus mechanisms on well-connected networks.

Parallelism Limits Across Network Topologies We evaluate performance across three common fixed network topologies while varying the number of machines: ring, 2D and 3D tori, and a 1-peer exponential graph (Ying et al., 2021). The exponential graph is a fast-mixing topology with spectral gap scaling as $1/\rho = \mathcal{O}(\log M)$.

In Figure 2, we plot the final error as a function of the number of machines across four configurations defined by

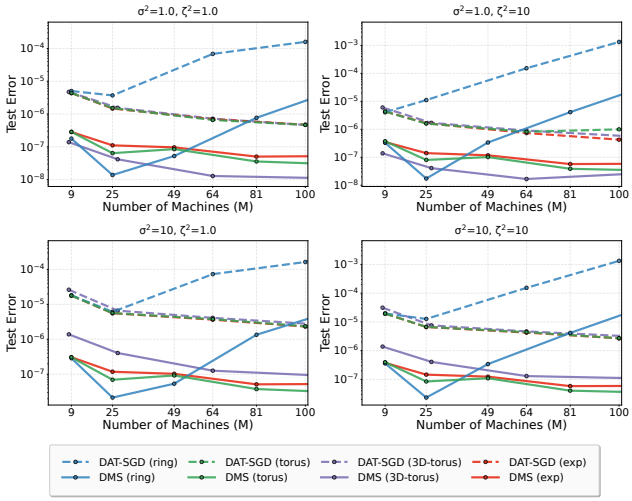

Figure 2. Test error on a synthetic least-squares problem across varying numbers of machines and different levels of noise (σ^2) and data heterogeneity (ζ^2), evaluated on ring, 2D and 3D tori, and exponential graph topologies. We report the averaged squared error $\frac{1}{M}\sum_{i=1}^{M}\|x_T^i - x^*\|^2$.

$\sigma^2, \zeta^2 \in \{1, 10\}$. For DAT-SGD (dashed lines), performance degrades sharply on the ring topology as M increases, while showing more stability on the torus and exponential graph topologies. In contrast, DMS (solid lines) maintains consistent or improving performance across all topologies: on the ring, DMS significantly outperforms DAT-SGD and remains stable up to $M = 25$ before showing slight degradation; on the torus and exponential graphs, DMS continues to improve or maintain low error even at $M = 100$.

These results demonstrate that DMS enables effective utilization of significantly more machines compared to DAT-SGD across all topologies, with the most noticeable improvements on poorly connected networks like the ring, thereby validating our improved parallelism scalability.

6. Discussion and Future Work

We introduced DMS , a decentralized optimization method that significantly expands the achievable parallelism in stochastic convex optimization. For a broad class of network topologies, DMS improves existing parallelism bounds and, for sufficiently well-connected graphs, matches the centralized scaling $\mathcal{O}(\sqrt{N})$, showing that the performance gap between centralized and decentralized learning is not fundamental.

We leave it for future work to extend the analysis to general mixing matrices beyond the symmetric and doubly stochastic setting (e.g., asymmetric or row/column stochastic), establish convergence guarantees for non-convex objectives, and characterize the maximal regime in which centralized-level parallelism can be achieved, including whether similar scalability extends beyond the convex setting.